Diego Alducin, Ph.D.

San Antonio, TX, USA diego.alducin@gmail.com (210) 291-5631 in/alducin-diego datdiego.github.io/

SUMMARY

Scientist and Engineer with a Ph.D. and a strong foundation in machine learning, AI, and advanced data analytics. Experienced in building and deploying scalable models for predictive analytics, anomaly detection, computer vision, and time-series forecasting across diverse domains including aerospace, automotive, cybersecurity, and semiconductors. Proven success in leading cross-functional teams, securing project funding, and delivering solutions that drive measurable impact. Skilled in Python, SQL, cloud computing (AWS/GCP), and modern ML frameworks. Adept at bridging the gap between research and application, with a passion for transforming complex data into actionable insights and intelligent products.

EXPERIENCE

Senior Computer Scientist

Southwest Research Institute **October 2024 - Present, San Antonio, TX**

- Developed an algorithm using object detection and advanced data analytics to identify specific events from the Juno Mission with 94% accuracy.
- Managing a $350k project with 3 engineers to investigate the effects of Electromagnetic Interference (EMI) on electric vehicles for the National Highway Traffic Safety Administration (NHTSA).
- Continually developing business relationships with customers to provide cutting-edge solutions, and leading efforts to obtain funding for new projects and technologies.
- Leading a team of 4 engineers to develop a new Large Language Model (LLM) to improve the efficiency of cybersecurity testing.

Research Computer Scientist

Southwest Research Institute **January 2023 - October 2024, San Antonio, TX**

- Implemented a secure micropatching algorithm for communications between space systems successfully deployed in the International Space Station (ISS) on an Amazon Web Services (AWS) Snowcone.
- Contributed to developing cutting-edge technology in the DARPA CRASH program.
- Developed a web application that uses React and Redux to create cybersecurity test plans for automotive systems.
- Led the creation of an API using Python to reverse engineer CAN bus messages from vehicles directly from the OBDII port with 91% accuracy.
- Created a Long short-term memory (LSTM) Neural Network that predicts the path and velocity of a vehicle based on Basic Safety Messages with 89% accuracy..

Expert Failure Analysis Engineer

Tower Semiconductor **May 2019 - January 2023, San Antonio, TX**

- Established the area of Failure Analysis within the fab to streamline requests between departments, improving analysis turnaround time by 80%.
- Created a ticketing database web system with SQL, JavaScript, PHP and Python that processed requests, set priorities, and collected data from analysis.
- Conducted in-depth failure analysis of diverse microelectronic device families in high-volume manufacturing environments using advanced diagnostic techniques and predictive analytics.
- Correlated failure mechanisms of technologies between electrical measurements, process development, and physical analysis using Python and KLA Data Analysis Environment.
- Involved in quality audits for customers to determine specific production process requirements.
- Negotiated and obtained capital, $600k, for state-of-the-art instruments and upgrades.
- Deployed cutting-edge failure analysis workflows, integrating advanced diagnostic techniques to boost issue resolution efficacy by 30% in the semiconductor sector.

Senior Process Engineer

Tower Semiconductor **December 2016 - May 2019, San Antonio, TX**

- Supported processes of 12 different technologies for large-scale production of 8-inch wafer semiconductors.
- Identified a cost-saving measure using classification algorithms which led to the introduction of a new tool to improve the lifetime of testing materials saving $750k within a year.
- Utilized high-level data analysis and regression algorithms to determine the contribution of process changes for chemical, physical, and environmental factors in products for yield improvement and technology transfers.
- Spearheaded a team in integrating data analytics platforms using Pandas and SQL, leading to a $1.4 Million reduction in test wafer costs.
- Developed new processes for technology transfers with SIPOC documentation.
- Improved process control above a Cpk of 1.67 of critical tools by introducing continuous VPD ICP-MS measurements to accurately detect impurities in chemical concentrations..

EDUCATION

M.Sc. Computer Science

University of Texas at San Antonio • 2024

- Coursework: Data Science, Artificial Intelligence, Machine Learning, Deep Learning, Software Engineering, Cloud Computing
- Final Project: "Physics-Informed Learning for Scalable Single-Qubit Control Simulations".

Ph.D. Physics

University of Texas at San Antonio • 2016

- Coursework: Holography, Materials Science, Nanotechnology, Semiconductor Physics, Instrumentation (Lab-View), Diffraction Physics, Mathematical and Computational Physics.
- Thesis Topic: "In situ Transmission Electron Microscopy Experimentation of Nanostructured Materials.".

B.Sc. Physics

University of Texas at San Antonio • 2012

SKILLS

Web Development & Cloud Technologies: HTML5, CSS, React, Flask, Docker, AWS, GCP.

Programming & Scripting: Python, C, C++, Java, JavaScript, SQL, Bash

Machine Learning & Data Science: Pandas, Numpy, PyTorch, TensorFlow, Keras, Scikit-learn, NLP, Computer Vision, Physics-Informed Neural Networks (PINNs), Object Detection, Anomaly Detection, Ollama, Huggingface, LLMs.

Data Engineering & Big Data: Apache Spark, Apache Airflow, Pandas, SQL.

Technical Documentation: Markdown, LaTeX, Microsoft Office (Word, Excel, PowerPoint, Access).
Languages: English (Native), Spanish (Native).